Mechanized Pseudomonas fluorescens Application via Aerosol to Combat White-nose Syndrome

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Project Categories: Field Conservation; Research

Abstract:

White-nose Syndrome (WNS) is a fungal infection caused by Pseudogymnoascus destructans (Lorch et al.) which has greatly reduced the populations of multiple bat species, such as the endangered Eastern small-footed bat, across North America. P. destructans is highly virulent and transmitted by physical contact, allowing it to quickly spread through a colony of bats Within five years from introduction, WNS can deplete up to 90% of a colony's numbers. (Langwig et al.) Because the fungus is cryophilic, it spreads during winter as the bats hibernate. The additional stress upon the bat's immune system causes them to rouse more frequently than is healthy, and they frequently starve.

No cure for WNS is known, but a bat's resistance to the disease can be supplemented. In a field test, the bacteria Pseudomonas fluorescens was found to increase a bat's survival rate by 500%. (Hoyt et al.)

Previous experiments with Ps. fluorescens have applied the bacteria via capturing individuals and swabbing them. Another method must be used for widespread protection. This project will explore the use of aerosolized Ps. fluorescens placed at high-traffic areas to treat more bats and reduce the effect of WNS. If successful, this project will reveal a method to significantly lower WNS fatalities. The additional surviving bats will consume pests that would otherwise damage crops or other plants, triggering a rise in pesticide usage.

Narrative:

Project Overview & Significance:

WNS is a threat to biodiversity across the United States directly and indirectly. Endangered species such as Myotis sodalis suffer from the disease, but it would be a grave concern even if it only affected the most common bats. If fewer bats live to consume pestilential insects, they will ravage crops and other plants in their area. The damage to crops is particularly concerning, as farmers will act as they see fit to protect their livelihoods. They will increase their pesticide use and indiscriminately harm wildlife nearby and downstream from their fields.

Goals, Objectives, and Actions:

This experiment hopes to provide a way conservation agencies can actively combat the bat dieoffs caused by WNS.

Researchers will strive to minimize disruption for the bats, both for their sake and the data's. They will keep their distance from the bats at all times besides tagging/trapping. All actions pertaining to the mister in the field will be performed in the evening, as the bats naturally rouse.

Methods:

This experiment will be performed using two bat colonies of similar density. The model species will be the cave myotis, as it is susceptible to WNS and has a relatively large population to select from. These colonies should also be in similar climates to ensure variations in winter temperature do not affect the results. At least one colony should be in a location with a single means of ingress for the bats, the narrower the better. This will be the experimental colony.

At each colony, researchers will use mist nets to trap bats, which will be checked for evidence of WNS. 30 infected individuals at each colony will be given radio tags and released.

Ps. fluorescens will be cultivated in a laboratory in bulk. The resulting suspension will be aerosolized in a mist of droplets roughly 450 micrometers in diameter, as this was found to maximize the survival of the related Ps. syringae. (Marthi et al.) A quiet electronic motor will be used to pump the suspension through a sprayhead over the experimental roost entrance. This will be done during phases of peak activity to maximise bat exposure while conserving the suspension. The bats will be misted for a week prior to the projected start of hibernation, as Ps. fluorescens has the greatest positive effect when administered after infection. (Cheng et al.)

Once officials familiar with the colonies believe all bats have ended hibernation, likely sometime in April, researchers will locate as many of the tagged bats as possible at both locations. They will repeat this in early May to assess long-term survival.

Timeline:

Note: Timings for bat interaction may vary from original plan, depending on recorded hibernation dates.

- Late July: Trial colonies selected and agreements made with the relevant local authorities. Mister design finalized.
- Second week of September: Mister built and placed near roost entrance.
- Third week of September:Ps. fluorescens suspension ready. Specimens, trapped, tagged, and released.
- First week of October: Ps. fluorescens suspension applied to bats at experimental location.
- Second week of April: Initial check on tagged bats.
- First week of May: Secondary check on tagged bats.

Evaluation & Outcomes:

For this project to be fully successful, the bats treated with Ps. fluorescens must have a significantly higher survival rate than the control group a month from the end of hibernation in May 2027. A temporary difference immediately following awakening will be considered a partial success, suggesting the technique may successfully improve WNS survival after refinement.

Dissemination:

The results of this experiment will be shared with the National Park Service, as well as any state park organization dealing with WNS or zoos with vulnerable species in captivity. News organizations focusing on bats, caving, or related topics will be notified as well.

Ethical and Welfare Considerations:

Experts in bat handling will be consulted on the optimal way to treat the bats during the tagging process, and how the misting device can be placed to minimize its disruption of the bats' activity.

Budget:

Item	Estimated Cost	Description
Equipment	$13,000	Radio tags and tracker. Mister. Mist net. Suspension Preparation.
Travel	$2,000	Gasoline/fuel.
Supplies	$5,000	Food, safety gear, printing.

Justification:

Travel is predicted to be relatively inexpensive for this operation, as cave myotis live nearby in Kansas. Supplies are mostly a low priority, as we predict little need for data processing or external communication. Regardless of the results, the general public does not need to modify their behavior to implement these lessons. We do not need to invest in active dissemination.

The radio tags and Ps. fluorescens are paradoxically crucial to the experiment and flexible. We need to have enough of each for a solid sample size and effective coverage, but the experiment and budget as written go beyond the minimum. In the event of partial funding, some money earmarked for those tools can be used elsewhere.

References:

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